

ECON 0100 | Demo B

Demos are similar to Checkpoints, often taken directly from past semesters. Work through the problems and check your work against mine. Practice answering clearly and completely. You are encouraged to work in small groups. Find a study room, grab some classmates, and work together on a whiteboard.

Dragontear

Dragontear is a rare and powerful substance used in potions, spells, and fire-breathing, harvested from dragons while they laugh uncontrollably. Prices are in coins and quantity is in vials. The supply and demand equations for dragontear are known to be the following:

$$P = 60 - \frac{1}{10}Q_D \quad \text{and} \quad P = \frac{1}{5}Q_S$$

Q1 | Demand and Consumer Surplus (B1.1, B1.2)

Use a graph to plot this demand curve. Find and label the consumer surplus at a price of 30. On the same graph, show what happens to consumer surplus when the price increases to 40.

- a) Quantity demanded at a price of 20 _____, at 30 _____, at 40 _____ (B1.1)
- b) What is the marginal benefit of the 300th vial? _____ (B1.1)
- c) CS at 30 _____ CS at 40 _____ Expenditure at 30 _____ (B1.2)



Q2 | Supply and Producer Surplus (B2.1, B2.2)

Use a graph to plot this supply curve. Find and label the producer surplus at a price of 30. On the same graph, show what happens to producer surplus when the price increases to 40.

- a) Quantity supplied at 20 _____, at 30 _____, at 40 _____ (B2.1)
- b) What is the marginal cost of the 100th vial? _____ (B2.1)
- c) PS at 30 _____ PS at 40 _____ Revenue at 30 _____ (B2.2)



Q3 | Price Elasticity (B5.2)

- a) Find the elasticity of demand when the price changes from 10 to 20. _____

Is this: *elastic*, *unit elastic*, or *inelastic*?

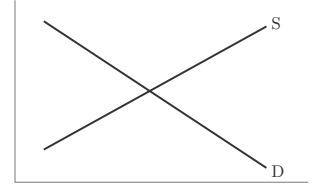
- b) Find the elasticity of supply when the price changes from 20 to 40. _____

Is this: *elastic*, *unit elastic*, or *inelastic*?

Q4 | Equilibrium for Dragontear (B3.1, B4.1)

Use a graph to describe the market for Dragontear:

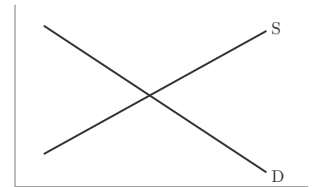
- a) Calculate equilibrium price: _____ Find equilibrium quantity: _____ (B3.1)
- b) At 30 coins: *shortage*, *excess* of size _____, and the price will: *rise*, *fall* (B3.1)
- c) Calculate total surplus at equilibrium: _____ (B4.1)
- d) What is the deadweight loss if only 100 vials are exchanged? _____ (B4.1)



Q5 | Price Floor (B4.2)

Use a graph of the demand curve and the supply curve to evaluate the welfare effects of a price floor of 45.

- a) CS _____ Change in CS from Equilibrium _____
- b) PS _____ Change in PS from Equilibrium _____
- c) DWL _____ Shortage/Excess _____



Q6 | Second Wizarding War (B5.1, B5.3)

The second wizarding war limited the population of dragons. Use a graph to aid your discussion of the impact the war had on the market for dragontear.

- a) The demand curve: *shifted up*, *stayed the same*, *shifted down* (B5.1)
- b) The supply curve: *shifted up*, *stayed the same*, *shifted down* (B5.1)

In the years after the war, many more in the wizarding world required dragontear as part of a widespread rise in the popularity of spells. On its own (not including the change in the population of dragons), how has this change in popularity of spells changed the market for dragontear?

- e) The demand curve: *shifted up*, *stayed the same*, *shifted down* (B5.1)
- f) The supply curve: *shifted up*, *stayed the same*, *shifted down* (B5.1)

Without using numbers, how have these two changes together impacted the dragontear market equilibrium?

- g) Equilibrium price has: *increased*, *stayed the same*, *decreased*, *indeterminate* (B5.3)
- h) Equilibrium quantity has: *increased*, *stayed the same*, *decreased*, *indeterminate* (B5.3)

